Playwright Auto-Waiting

Reference : https://playwright.dev/docs/actionability

- **Auto-waiting** is a built-in Playwright feature that automatically waits for an element to become ready before performing an action.
- Before executing actions (such as `click()`, `fill()`, or `check()`), Playwright performs **actionability checks** to ensure the element is ready for interaction.
- Playwright executes the action **only after all required checks pass**.
- If the required conditions are **not met within the specified timeout**, Playwright throws a `TimeoutError`.
- Auto-waiting helps make tests **more reliable** and **reduces flaky test failures**.
- It eliminates the need for most **manual waits** like `waitForTimeout()`.

Actionability Checks

Before performing an action, Playwright verifies that the element is:

- ✅ Present in the DOM
- ✅ Visible
- ✅ Stable (not moving or animating)
- ✅ Enabled
- ✅ Able to receive user events (not covered by another element)

Forcing Actions

- Some actions (such as `locator.click()`) support the `force: true` option.
- Using `force: true` skips certain **non-essential actionability checks**.
- Example:

```
await page.locator('#submit').click({ force: true });
```

- Use `force` **only when necessary**, as it may interact with elements that are not actually clickable by a real user.

Auto-Retrying Assertions

- Playwright assertions automatically **retry** until the expected condition becomes true or the timeout is reached.
- This behavior is similar to auto-waiting for actions.
- Auto-retrying assertions make tests **more stable** and reduce intermittent failures.

Key Benefits

- ✔ No need for explicit waits in most cases.
- ✔ Reduces flaky tests.
- ✔ Improves test reliability.
- ✔ Simulates real user interactions more accurately.
- ✔ Simplifies automation code by handling waits automatically.